FIRST SEMESTER SESSION 2025/2026 A251

SQQPK3033 COMPUTER MODELING IN BUSINESS

GROUP A

GROUP ASSIGNMENT 2

EXPAND THE DES MODEL (PHASE 2)

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Executive Summary

This study presents an expanded Discrete Event Simulation (DES) model of Teja Fresh Mart (TFM), focusing on the fresh produce, butchery, and payment counter sections to analyse operational performance. The model incorporates multiple customer types, scheduled arrivals for peak and off-peak hours, staff scheduling, cycle time tracking, and full entity transfer animations, providing a realistic representation of store operations. A What-If analysis was conducted by maintaining a steady workforce of two staff members across all sections and compared against the Basic Model. The results show that the What-If Model served more customers while reducing both average total time in the system and average waiting time, indicating improved service efficiency. Maximum waiting times decreased, and staff utilization increased moderately, reflecting balanced resource engagement without overworking employees. Despite a slight increase in lost customers during peak periods, the analysis demonstrates that consistent staffing can enhance overall system performance. It is recommended that TFM adopt steady staffing during peak hours and explore dynamic allocation strategies based on real-time demand, while regularly monitoring key performance metrics to ensure continued operational effectiveness.

1.0 Introduction

Teja Fresh Mart (TFM), located in Bukit Kayu Putih, is a retail store offering fresh produce, bakery, household items, butchery, deli, and dairy. Among these, the fresh produce, butchery, and payment counter sections experience the highest customer traffic, making them critical for operational analysis.

Discrete Event Simulation (DES) provides a structured approach to representing complex systems where changes occur at discrete events, such as customer arrivals or service completions. The initial DES model at TFM represented basic customer flow, including arrivals, service at the fresh produce and butchery sections, and processing at the payment counter.

Building on this foundation, the expanded DES model incorporates:

- Multiple entity types representing different customer categories,
- Scheduled arrivals for peak and non-peak hours,
- Resource reallocation and worker scheduling,
- Cycle time tracking across stations, and
- Full entity transfer animations.

These enhancements allow the model to represent store operations more realistically and provide a foundation for detailed analysis of customer flow, resource usage, and cycle times. The expanded DES model also supports scenario testing and what-if analyses, enabling the examination of different staffing or scheduling strategies without implying that operational efficiency is being improved in this stage.

2.0 Problem Justification

The initial Discrete Event Simulation (DES) model of Teja Fresh Mart (TFM) represents basic customer flow through the fresh produce, butchery, and payment counter sections. However, it has several limitations that reduce its ability to capture store operations fully.

The model includes only a single type of customer entity, does not account for scheduled arrivals during peak and non-peak hours, and simplifies resource allocation and worker scheduling. In addition, cycle times across stations are not tracked, and entity transfer animations are incomplete, limiting the visualization of customer flow.

Expanding the DES model to incorporate multiple entity types, scheduled arrivals, resource reallocation, worker scheduling, cycle time tracking, and full animations allows for a more realistic representation of store operations.

This enhanced model provides a foundation for detailed analyses, such as scenario testing and evaluation of different operational setups, aligning with the assignment requirement to expand the business system rather than directly improving performance.

3.0 Objective of the study

The objective of this study is to expand the Discrete Event Simulation (DES) model for Teja Fresh Mart (TFM) to better represent its operational system. The following specific actions support this objective:

1. To incorporate multiple entity types representing different customer categories in the model.
2. To implement scheduling-based arrivals for peak and non-peak hours.
3. To simulate resource reallocation and worker scheduling across different stations.
4. To track cycle times for customers across the fresh produce, butchery, and payment counter sections.
5. To implement full entity transfer animations for improved visualization of customer flow.
6. To provide a detailed representation of store operations that supports future scenario testing and system analysis.

4.0 Scope and Significance of the Study

This study focuses on the fresh produce, butchery, and payment counter sections of Teja Fresh Mart (TFM). The analysis considers customer arrivals, service processes, and staff allocation over a defined period, tracking cycle times, resource allocation, and entity flow. Other store sections, such as bakery, dairy, or pricing strategies, are excluded.

The significance of this study lies in providing a more detailed and realistic simulation of TFM's operations. By expanding the model, management and analysts can visualize customer flow, analyse resource allocation, and test alternative operational scenarios. The expanded DES model serves as a tool for understanding system dynamics and supporting informed decision-making regarding operational planning and resource management.

5.0 Data interpretation

The data for the expanded model were interpreted to reflect a more realistic environment over a 3-day (4,320 minutes) duration.

- Arrival Patterns: Instead of constant arrivals, data was categorized into an Arrival Schedule to distinguish between Peak (high traffic) and Off-Peak (low traffic) periods.
- Staffing Levels: Data reflects a Resource Schedule where staff capacity changes according to shifts rather than remaining fixed.
- Failure/Re-entry Data: Statistical data assumes a 5% dissatisfaction rate, where customers enter a re-entry loop to restart their process.
- Service Times: Cycle times are tracked specifically for the Fresh Produce and Butchery stations to identify station-specific delays.

6.0 Model assumptions

To ensure the simulation reflects realistic store behaviour, the following assumptions are applied:

- Operating Hours: The store follows a 24-hour cycle where certain blocks (e.g., Night) have an arrival rate of 0 or significantly lower.
- Balking Logic: If queue capacities are exceeded, it is assumed customers will leave the system (balk) rather than wait indefinitely.
- Re-entry Probability: A 5% "Unsatisfied" rate is assumed, where customers must restart the process through a re-entry loop.
- Entity Transfer: It is assumed that transfer times between stations (Fresh Produce → Butchery → Payment) are consistent with the store's physical layout.

7.0 Verification & Validation process

- Verification:

The model was verified by checking the Schedule Module to ensure the arrival values and durations (totalling 24 hours per cycle) correctly repeated over the 3-day run period. Additionally, entity flow, resource allocation, and process logic were examined to confirm that the model behaves as intended without errors.

- Validation:

The GA2 Expanded Model was validated by comparing its performance with the GA1 Basic Model. While GA1 only simulated up to 120 customers, GA2 demonstrates that the system can handle variable customer loads over a full 4,320-minute simulation period without crashes or infinite queue growth. Key performance measures, including customer throughput, queue lengths, and staff utilization, were monitored and verified. These results confirm that the GA2 model realistically represents the operational processes and can reliably simulate customer flow and staff interactions across different sections.

8.0 Analysis of the basic model

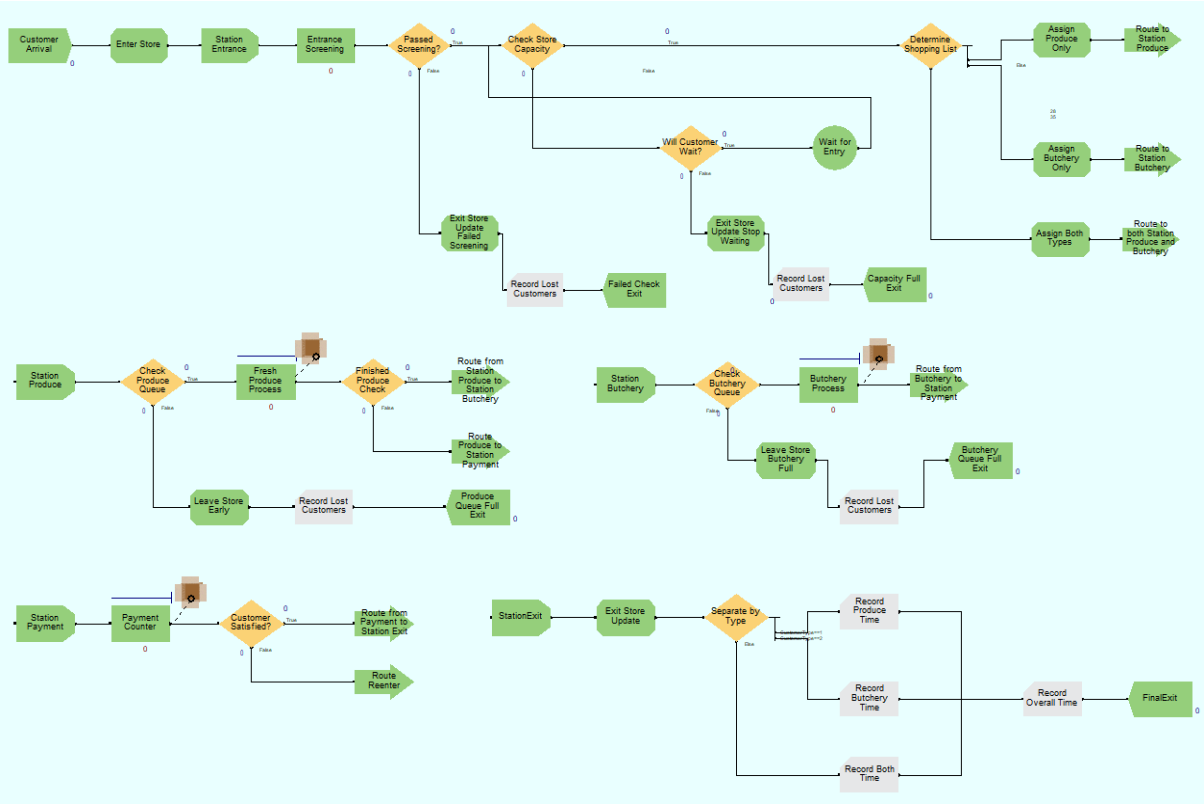


Figure 1: Flowchart of Basic Model

The initial simulation of Teja Fresh Mart (TFM) represents a flexible staffing schedule with variable worker allocation according to customer arrivals throughout the 24-hour cycle. This model serves as the baseline to observe system behaviour under normal operating conditions.

Based on the simulation results, the following observations are made for the Basic Model:

- Queue Length: The Payment Counter.Queue.NumberInQueue shows moderate queueing, with occasional peaks during busy hours. The fresh produce and butchery sections maintain low queue lengths due to staggered arrivals and sufficient service capacity.
- Utilization: Scheduled Utilization for workers is relatively low, averaging around 46%, indicating that staff in fresh produce and butchery sections are underutilized, while the payment counter experiences occasional higher utilization.
- Wait Time Spikes: Waiting times remain generally stable, but there are intermittent peaks during busy periods. The half-width of waiting times is smaller compared to the fixed-resource scenario, suggesting more predictable service and minimal extreme delays.

9.0 What-if analysis (Scenario: Fixed Amount of Resources)

The experimentation phase for Teja Fresh Mart (TFM) involves a Resource Sensitivity Analysis. The primary goal is to determine if the store can maintain operational efficiency while simplifying labour management by moving from a flexible schedule to a fixed staffing level. In this experiment, the Worker resource capacity is set to a fixed value (2 workers) that remains constant for the entire 24-hour cycle, regardless of the arrival schedule.

Based on the high maximum wait times in the Basic Model, the following results from this What-if analysis:

- Queue Length: Significant growth in the Payment Counter.Queue.NumberInQueue during the specific hours where the arrival schedule exceeds 15 customers per hour.
- Utilization: An increase in Scheduled Utilization from the current approximate 46% toward 80-90%, representing a more "stretched" workforce.
- Wait Time Spikes: A likely increase in the Half Width of waiting times, indicating that the system becomes less predictable and more prone to extreme delays when resources are fixed.

10.0 Comparison of basic & experimentation model

The table below illustrates the changes in key metrics that compares the Basic Model to the What-if Analysis, in which a steady workforce of two staff members is maintained across all sections.

Metrics	Basic Model	What-if Analysis	Change
Customers Served	364.00	373.80	+9.8
Average Total Time in System	19.56 minutes	17.16 minutes	-2.4 minutes
Average Waiting Time	4.62 minutes	2.11 minutes	-2.51 minutes
Maximum Waiting Time	746.16 minutes	716.07 minutes	-30.09 minutes
Cashier Utilization	45.86%	47.71%	+1.85%
Fresh Produce Utilization	20.95%	21.89%	+0.94%
Butchery Utilization	23.39%	23.79%	+0.40%
Lost Customers	16.8 ≈ 17	18.2 ≈ 18	+1

Table 1: Comparison between Basic Model and What-if Analysis

Based on the results in Table 1, the What-If Model demonstrates improved performance compared to the Basic Model. The number of customers served increased, while both the average total time in the system and the average waiting time across all queues decreased. This indicates that the system processed customers more efficiently and reduced overall delays.

The reduction in total and waiting times is likely due to more consistent staff availability, which allowed service points to operate more smoothly and handle customer demand effectively. Additionally, the decrease in maximum waiting time shows that extreme delays were minimized, improving customer experience.

Staff utilization rates for cashiers, fresh produce, and butchery increased moderately, suggesting that employees were more effectively engaged without being overworked. Despite these improvements, there was a small increase in lost customers, possibly due to temporary congestion during peak periods.

Overall, the What-If Model highlights that maintaining a steady workforce can enhance system efficiency, reduce waiting times, and better balance staff workload, leading to improved overall performance.

11.0 Conclusion and recommendation

In conclusion, the expanded DES model for Teja Fresh Mart demonstrates that staffing levels significantly impact operational efficiency. The comparison of the Basic Model with the What-If Model shows that maintaining two staff members consistently across all sections improves customer flow and service speed, evidenced by higher throughput and reduced waiting times. Maximum delays were minimized, enhancing customer experience, while staff utilization improved without overburdening employees. However, the customer waiting time and balking rate reveal that the system becomes unstable during peak periods, with extreme maximum waits of 716.07 minutes, indicating that a fixed staffing level of one or two cashiers is insufficient to clear the accumulated queue.

Based on these insights, TFM should reject the fixed staffing approach and continue using a variable scheduling system that aligns staff levels with arrival patterns. Additionally, dynamic staff reallocation is recommended: employees in low-utilization sections, such as butchery and fresh produce, can be temporarily assigned to the payment counter when its queue exceeds four customers. To further reduce lost customers, buffer capacity improvements should be considered, including expanding payment queue space or implementing self-service kiosks to handle overflow during peak periods. By adopting these strategies, TFM can maintain consistent service efficiency, optimize workforce utilization, and enhance overall customer experience.